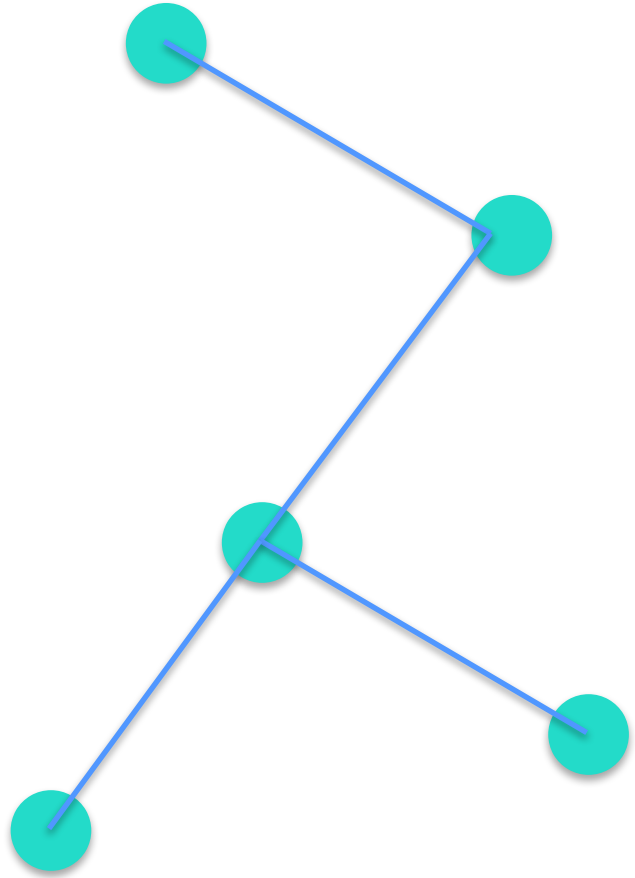
Privacy-Aware IPv6 IoT Security Monitoring

Monitoring security events across temporary IPv6 address rotation



HACKHIVE

Team code: A3-PS007-TC342

BMS College of Engineering

Problem statement: PS-007

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Executive summary

- IoT devices can rotate temporary IPv6 addresses, making event correlation harder for a local defender.
- This project proposes a privacy-aware monitoring pipeline for an isolated IPv6 IoT LAN.
- Short-lived local pseudonyms and authenticated sessions support limited continuity across address changes.
- Detection focuses on rogue Router Advertisements (RAs) and abnormal Neighbor Discovery Protocol (NDP) rates.

KEY POINT

Outcome: useful local security alerts with deliberately bounded correlation.

PROPOSED • NOT YET VALIDATED

Problem context: IPv6 address rotation

- IPv6 privacy extensions can generate temporary addresses to reduce long-term address linkability.
- A device’s address may change while its local security session remains valid.
- Address-only monitoring can fragment one device’s events across multiple addresses.
- Permanent device identifiers would make correlation easier but increase privacy risk.

KEY POINT

Design principle: correlate only when needed, only locally, and only for a bounded period.

PROPOSED • NOT YET VALIDATED

Threat model and assumptions

- Environment: one isolated IPv6 LAN with five emulated IoT devices and a controlled router.
- Defender can observe local RA/NDP traffic and administer the test environment.
- Potential threats include an unauthorised RA source and an abnormal volume of NDP messages.
- Address rotation is expected behaviour, not itself a malicious event.

KEY POINT

All attack simulations should remain inside the authorised lab environment.

PROPOSED • NOT YET VALIDATED

Project objectives

- Create an IPv6 testbed with five emulated devices and configurable address rotation.
- Implement authenticated local sessions and short-lived pseudonym assignment.
- Detect rogue or unauthorised Router Advertisements using a defined local policy.
- Detect abnormal NDP rates using configurable thresholds and a benign baseline.
- Measure detection rate, false-positive rate, alert latency, and processing overhead.

KEY POINT

Success will be judged using repeatable tests—not unsupported claims.

PROPOSED • NOT YET VALIDATED

Proposed solution at a glance

Collect local IPv6 security events.

Associate events with a temporary local pseudonym through an authenticated session.

Apply rule-based RA validation and rate-based NDP anomaly detection.

Generate alerts with reason, time, and minimum necessary context.

Expire pseudonym mappings and periodically verify retention behaviour.

The monitor should not require a permanent hardware identity.

System architecture



Privacy guardrails across every module: local-only correlation • short retention • minimum event data • no hardware fingerprinting

Architecture shown is proposed; implementation details may change during prototyping.

Module 1 — IPv6 test environment

- Create one isolated IPv6 LAN with a router and five emulated devices.
- Configure temporary IPv6 address rotation at controlled intervals.
- Generate normal baseline traffic before introducing any attack scenario.
- Record test parameters such as rotation interval, traffic rate, and trial duration.

KEY POINT

Evidence to capture: topology diagram, configuration summary, and repeatable test procedure.

PROPOSED • NOT YET VALIDATED

Module 2 — Session and pseudonym manager

- Enrol each emulated device within the local testbed.
- Establish an authenticated session before associating events with a pseudonym.
- Issue a random, short-lived local pseudonym rather than exposing a permanent device identifier in routine logs.
- Permit limited session continuity across address rotation only while the session remains valid.

KEY POINT

Privacy boundary: no persistent hardware fingerprinting or cross-tenant mapping.

PROPOSED • NOT YET VALIDATED

Module 3 — Rogue Router Advertisement detection

- Define which router/interface is authorised to send Router Advertisements in the lab.
- Observe RA source and relevant message metadata on the local link.
- Flag RAs that violate the configured authorised-source policy.
- Include alert reason and timestamp so a reviewer can reproduce the event.

KEY POINT

A detection alert indicates a policy violation; validate assumptions before treating it as a real-world incident.

PROPOSED • NOT YET VALIDATED

Module 4 — Abnormal NDP rate detection

- Count selected NDP message types over a configurable sliding time window.
- Establish a benign baseline under ordinary device activity and address rotation.
- Trigger an alert when rate or burst thresholds are exceeded.
- Tune thresholds against benign trials to reduce false positives.

KEY POINT

Thresholds should be reported with the test conditions that produced them.

PROPOSED • NOT YET VALIDATED

End-to-end workflow

1. Start the isolated IPv6 LAN and confirm baseline connectivity.

2. Enrol devices and establish authenticated local sessions.

3. Assign temporary pseudonyms and enable address rotation.

4. Collect RA/NDP events and apply configured detection rules.

5. Generate controlled attack scenarios and record alerts.

6. Check alert accuracy, expire pseudonym mappings, and inspect retained logs.

Repeat each scenario across multiple trials for comparable results.

Attack scenario A — Rogue RA

- Goal: verify the monitor flags an unauthorised Router Advertisement source.
- Preparation: configure the authorised router policy and confirm normal RAs are accepted.
- Test: introduce a controlled unauthorised RA source in the isolated testbed.
- Expected observation: alert identifies a policy violation and includes timestamp/reason.

KEY POINT

Expected behaviour is a test hypothesis until demonstrated by implementation.

PROPOSED • NOT YET VALIDATED

Attack scenario B — NDP burst / flood

- Goal: identify abnormal NDP message volume without flagging ordinary traffic excessively.
- Preparation: measure benign NDP activity during normal device operation and rotation.
- Test: generate a controlled burst of NDP messages in the isolated LAN.
- Expected observation: configured rate threshold triggers an alert within the test window.

KEY POINT

Do not run flood simulations on public, campus, or production networks.

PROPOSED • NOT YET VALIDATED

Evaluation methodology

- Use repeated benign and attack trials with documented test conditions.
- Detection rate = true-positive attack trials ÷ total attack trials.
- False-positive rate = benign trials incorrectly alerted ÷ total benign trials.
- Alert latency = time from attack start/trigger to monitor alert.
- Measure resource overhead where feasible (CPU, memory, event-processing delay).

KEY POINT

Do not insert numerical results until the tests have actually been run.

PROPOSED • NOT YET VALIDATED

Privacy and data handling

- Local-only mapping: pseudonym-to-session association stays inside the test LAN.
- Short retention: remove expired pseudonym mappings according to a documented TTL.
- Data minimisation: log only event fields needed for detection and evaluation.
- No permanent identity recovery, hardware fingerprinting, or cross-tenant correlation.
- Review whether logs can indirectly relink events after expiry; document residual risks.

KEY POINT

Security monitoring and privacy protection must be evaluated together.

PROPOSED • NOT YET VALIDATED

Risks, limitations, and mitigations

- False positives: tune thresholds using benign baseline trials and report the chosen settings.
- False negatives: test varied attack rates and acknowledge coverage limits.
- Session compromise: use authenticated sessions and restrict mapping access.
- Over-retention: enforce expiry and test deletion behaviour.
- Lab-to-production gap: clearly label results as testbed-specific.

KEY POINT

A transparent limitations section improves technical credibility.

PROPOSED • NOT YET VALIDATED

Implementation roadmap

Phase 1: repository setup, team roles, and detailed design.

Phase 2: build the isolated IPv6 LAN and five-device emulation.

Phase 3: implement session/pseudonym lifecycle and local event collection.

Phase 4: add RA policy detection and NDP rate detection.

Phase 5: run baseline and attack trials; calculate evaluation metrics.

Phase 6: complete privacy analysis, technical report, and demo recording.

Update this roadmap as work is completed; do not mark planned tasks as done.

Deliverables

- Monitoring pipeline for local IPv6 RA/NDP events.
- Pseudonym and authenticated-session manager with expiry behaviour.
- Address-rotation test environment with five emulated devices.
- Controlled attack simulation and detection module.
- Evaluation report: detection rate, false positives, latency, and overhead.
- Privacy analysis covering mapping expiry and correlation risk.

KEY POINT

Deliverables are planned outputs, not claims of current completion.

PROPOSED • NOT YET VALIDATED

Team and responsibilities

- Team: HackHive | Team code: A3-PS007-TC342
- Institute: BMS College of Engineering
- Team member 1: Bhagya | USN: 1BF25AI035
- Team member 2: Deekshitha | USN: 1BF25AI040
- Mentor (if applicable): [Add name]

KEY POINT

Confirm team roles and replace any remaining placeholders before submission.

PROPOSED • NOT YET VALIDATED

Conclusion and next steps

- Build a local IPv6 monitor that remains useful as temporary addresses rotate.
- Combine short-lived pseudonyms and authenticated sessions with RA/NDP detection.
- Evaluate accuracy and latency using repeatable benign and attack trials.
- Test expiry and local-only correlation, and openly document residual privacy risk.

KEY POINT

HackHive • Privacy-aware security monitoring for IPv6 IoT

PROPOSED • NOT YET VALIDATED